

A comprehensive review on natural phenolic compounds as alternatives to in-feed antibiotics

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Intensive livestock and poultry farming in China largely relied on the use of in-feed antibiotics until July 2020. The consequences of antibiotic overuse in animal feed include accumulation in animal products and the development of bacterial antibiotic resistance, both of which threaten food safety and human health. China has now completely banned the circulation of commercial feed containing growth-promoting drug additives (except Chinese herbal medicine). Therefore, alternatives to in-feed antibiotics in animal production are greatly needed. Natural phenolic compounds (NPCs) exist widely in plants and are non-toxic, non-polluting, highly reproducible, and leave little residue. Many natural flavonoids, phenolic acids, lignans, and stilbenes have polyphenol chemical structures and exhibit great potential as alternatives to antibiotics. In this review we delineate the characteristics of plant-derived NPCs and summarize their current applications as alternatives to in-feed antibiotics, aiming to provide new strategies for antibiotic-free feeding and promote the development of more sustainable animal husbandry practices.

natural phenolic compounds, alternatives to in-feed antibiotics, animal husbandry, intestinal health

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Introduction

In-feed antibiotic supplementation in livestock and poultry production provided numerous benefits in previous decades, such as the inhibition of pathogenic intestinal bacteria, disease prevention, reduced mortality, enhanced growth, improvement of feed conversion rate, reduced farming costs, and shortening of the slaughter cycle (Castanon, 2007; van den Bogaard and Stobberingh, 1996). Unfortunately, this practice increased the prevalence of antibiotic-resistant pathogens in both humans and animals, posing a significant public health threat and impeding modern animal husbandry

(Giedraitienė et al., 2011; Kümmerer, 2009). Consequently, China's Ministry of Agriculture and Rural Affairs issued Notice No. 194, banning commercial feed containing growth-promoting drug additives (except Chinese herbal medicines) beginning 1 July 2020, marking the arrival of the “antibiotic-free era” in China's agriculture. Therefore, breeding industry desperately needs low-cost and effective alternatives to in-feed antibiotics for ensuring the long-term sustainability of China's livestock and poultry industry.

Such strategies must improve animal intestinal health and shape a healthy intestinal microbiome to be successful. Probiotics, prebiotics, enzyme preparations, acidifiers, antimicrobial peptides, bacteriophages, and plant active ingredients all comprise promising alternatives to in-feed antibiotics. The European Union's experience with anti-

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biotic-free farming in previous decades has proven that in-feed antibiotic replacement necessitates a comprehensive strategy rather than a single product. Consequently, natural phenolic compounds (NPCs), natural bioactive compounds derived from plants, have gained attention for their potential to improve animal growth and health in this context. Due to their antibacterial (Bai et al., 2016), anti-inflammatory (Li et al., 2018), antioxidative (Xu et al., 2016), immune regulatory (Gao et al., 2001), and intestinal health-promoting effects (Yuan et al., 2018), NPCs (e.g., flavonoids, chlorogenic acid, and curcumin) are gradually gaining acceptance and application in China's farming industry. In this review, we introduce the general properties of NPCs and summarize their current research applications to assess NPCs as alternatives to in-feed antibiotics.

Classifications of NPCs

NPCs are a class of compounds existing in plants that contain polyphenol structures. They are abundant in vegetables, fruits, and cereals. In plants, polyphenol content is surpassed only by cellulose, hemicellulose, and lignin (Brglez Mojzer et al., 2016). Major classes of NPCs include flavonoids, phenolic acids, lignans, and stilbenes (Figure 1) (Zhou et al., 2016).

Flavonoids

The basic structural feature of flavonoid compounds is the 2-phenyl-benzo [α] pyrane or flavane nucleus, which consists of two benzene rings (A and B rings) linked through a het-

erocyclic pyrane ring (C rings) (Figure 1, Flavone core) (Kumar and Pandey, 2013). Flavonoids are major secondary metabolites derived from the plant phenylpropanoid pathway (Wang et al., 2020c). Generally, the uptake of NPCs in-feed is limited by their low bioavailability and poor water solubility. However, the glycosylated forms of flavonoids that exist as glycosides in plant epidermal cell vacuoles have much better solubility, stability, and bioavailability. These advantages make flavonoids easier to extract and utilize (Sordon et al., 2016; van Genderen and van Hemert, 1986). Depending on the degree of oxidation, ring structure, and B ring position to the C-2 or C-3 of the central three-carbon storms, flavonoids can be further classified into flavones, isoflavones, flavonols, flavan-3-ols, flavonones, chalcones, and anthocyanidins (Figure 2). Additionally, the structural diversity of flavonoids is determined by the specific hydroxylation, methoxylation, and glycosylation of the benzene rings (Cushnie and Lamb, 2005). The abundance of flavonoids in plants is species-specific. For example, flavonols are commonly found in fruits, vegetables, and teas, whereas isoflavones are almost exclusively found in the legume family (Beecher, 2003).

To date, more than 7,000 structural variants have been identified in the flavonoid family (Nabavi et al., 2020). Bioactive properties of flavonoids have been widely reported, with antioxidant capacity being the best-known, which is affected by the number and position of hydroxyl groups on the A and B rings as well as the bond position between the B and the C rings. The key factors determining the bioactivity of flavonoids are, in descending order: the catechol structure on the benzene ring (especially 3',4'-OH), the existence of the 3-OH group on the C ring and the 5-OH

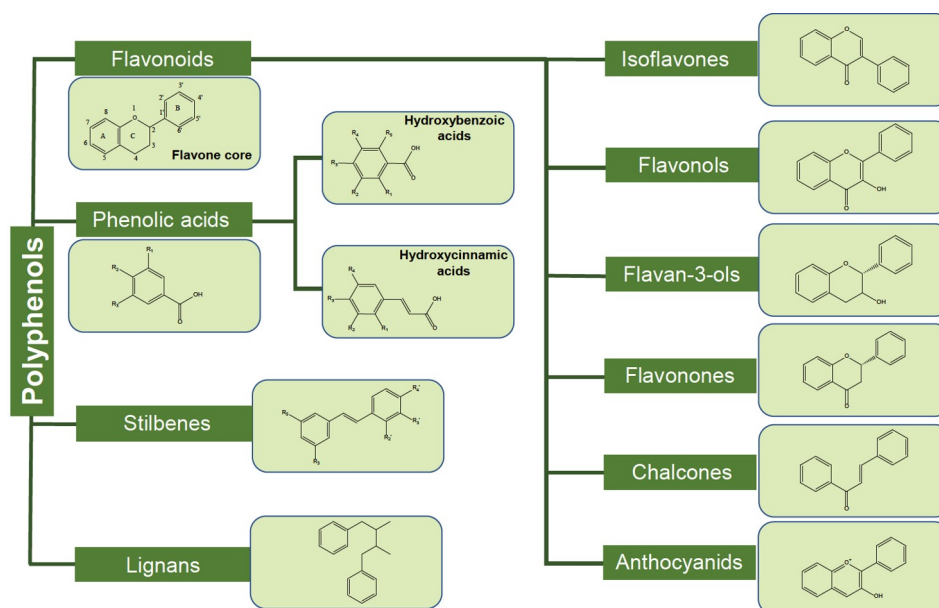


Figure 1 Chemical structures and classification of representative natural phenolic compounds (polyphenols).

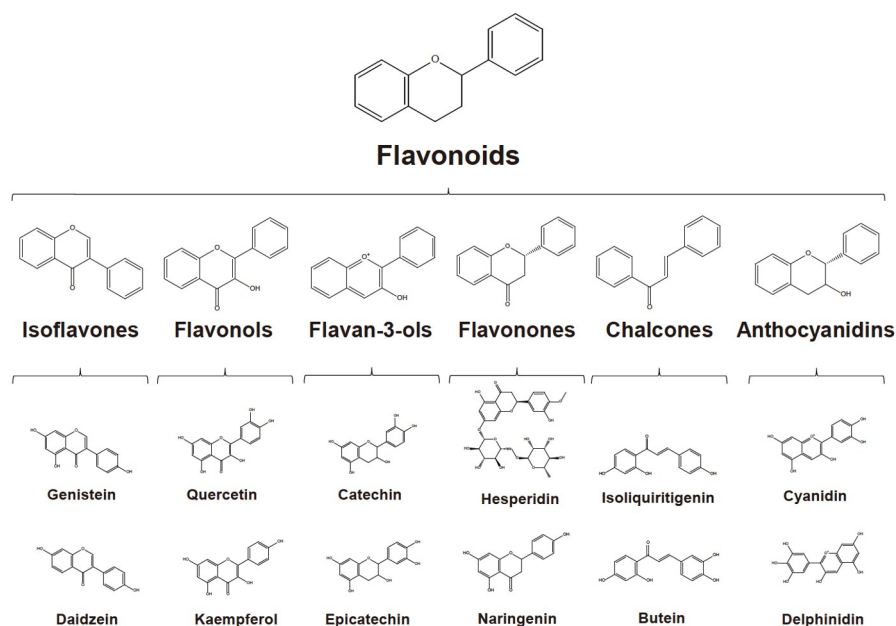


Figure 2 Chemical structures and classification of representative flavonoids: isoflavones, flavonols, flavan-3-ols, flavonones, chalcones, and anthocyanidins.

group on the A ring, and the structure of C2-C3 double bond conjugated with the 4-carbonyl group (Spiegel et al., 2020). For example, quercetin and its derivatives (e.g., rutin, quercetin, and hyperin) contain all the above and exhibit considerable antioxidant capacity (Lesjak et al., 2018).

Phenolic acids

Phenolic acids are composed of a carboxyl group and one or more hydroxyl groups bound to an aromatic ring. They exist in a variety of plant materials, especially seeds, fruit skins, and vegetable leaves (Karasawa and Mohan, 2018). Phenolic acids exist as free small molecules, except a small proportion in the polymer forms of tannic acid and lignin. At present, more than 8,000 natural phenolic acids have been identified (Yao et al., 2021). Like flavonoids, their bioactivity is modulated by hydroxyl and methoxyl groups on their aromatic rings (Valanciene et al., 2020). In contrast to flavonoids, which are mainly absorbed by microbiota, free phenolic acids (such as benzoic acid, phenylacetic acid, and cinnamic acid) can be directly absorbed by the stomach due to their high bioavailability and water/lipid solubility (Barrington et al., 2009; Kahle et al., 2011). Phenolic acids are typically classified into hydroxybenzoic acids and hydroxycinnamic acids (Figure 3) (Clifford, 1999). Hydroxybenzoic acids have seven-carbon cores, whose diversity is further dominated by various modifications on their aromatic ring. Representative hydroxybenzoic acids include gallic acid, ellagic acid, protocatechuic acid, syringic acid, p-hydroxybenzoic acid, salicylic acid, gentisic acid, and vanillic

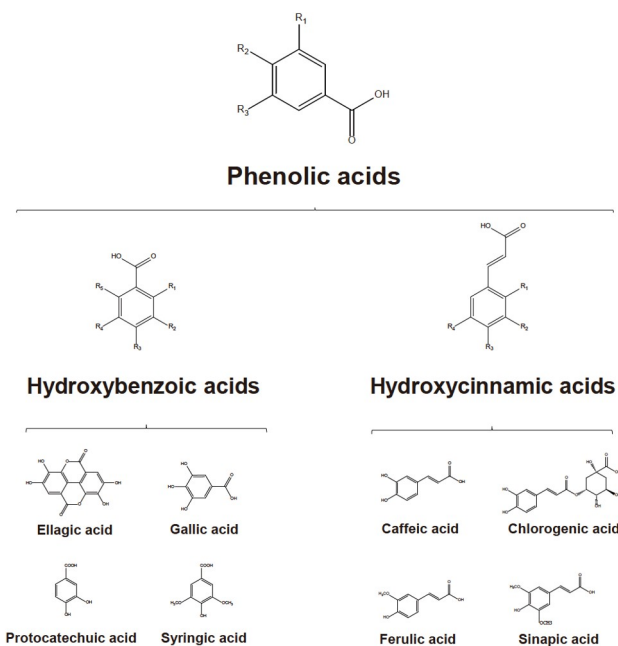


Figure 3 Chemical structures and classification of the representative phenolic acids: hydroxybenzoic acids and hydroxycinnamic acids.

acid. Hydroxycinnamic acids are phenolic compounds with a nucleus of cinnamic acid and a C6-C3 structure. They are more common in nature than hydroxybenzoic acids and usually exist in various conjugated forms. Common hydroxycinnamic acids include curcumin, coumaric acid, cinnamic acid, caffeic acid, ferulic acid, erucic acid, iso-ferulic acid, and p-hydroxycinnamic acid (Rashmi and Negi, 2020).

Stilbenes and lignans

Stilbene compounds are characterized by 1,2-diphenylethylenic nuclei (Sirerol et al., 2016) and mainly exist in grapes, wine, cranberries, strawberries, and peanuts (Ozdalet al., 2016). *Trans*-stilbenes are common in plants (Figure 4) and are generally antibacterial (Prukała and Kędzia, 1999), anti-inflammatory (Gao et al., 2006), and antioxidative (Cai et al., 2003). For example, *trans*-resveratrol is a typical stilbene compound that exhibits multiple bioactivities properties that promote animal health (Pecyna et al., 2020).

Lignans, a class of natural plant compounds (Figure 5) in

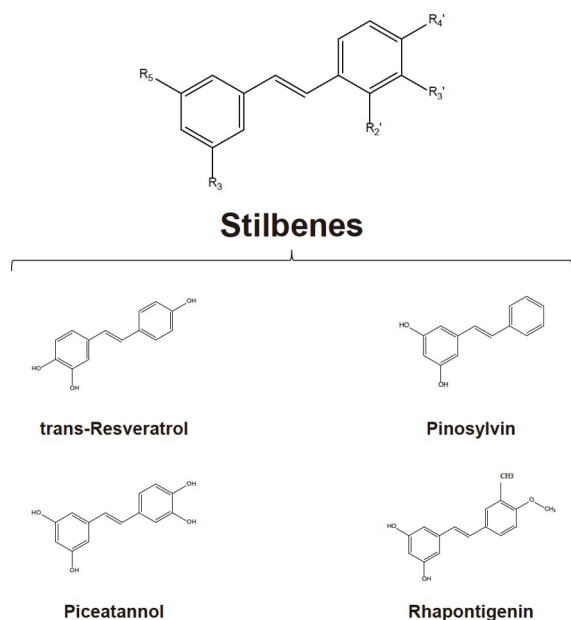


Figure 4 Chemical structures and classification of representative stilbenes.

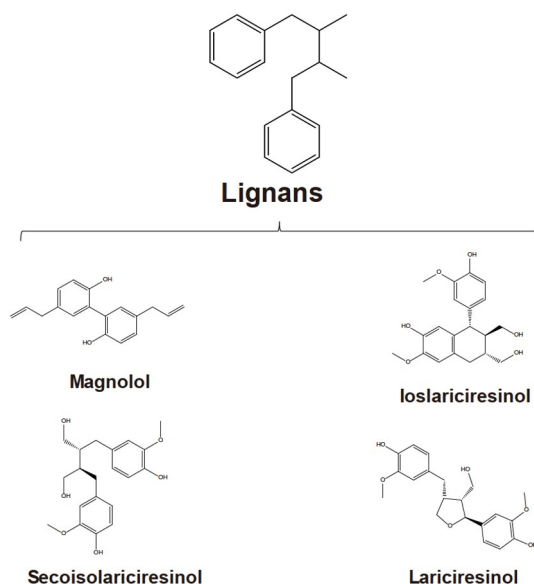


Figure 5 Chemical structures and classification of representative lignans.

flaxseed, sesame, and pumpkin, are commonly synthesized by oxidative polymerization of two or more phenyl propyl units (Cui et al., 2020a). Monomers of lignans include cinnamic acid, cinnamyl alcohol, propylene, and allyl benzene (Khoddami et al., 2013). Complex lignans include magnolol, isolariciresinol, hydroxymatairesinol, secoisolariciresinol, lariciresinol, pinoresinol, and matairesinol (Nørskov et al., 2015).

The bioactive properties of NPCs

NPCs have many beneficial effects on animal health. Studies have demonstrated their bioactive properties include regulating microflora, antioxidative effects, antibacterial and antiviral properties, immune regulation, and promoting intestinal epithelial barrier maintenance (Figure 6).

NPCs regulate intestinal microbiota

Composition of intestinal microbiota is strongly associated with the occurrence and development of various diseases (Ding et al., 2021). Additionally, the interactions between the commensal microbiota and viral infection of the farmed animals were verified by studies (Ran et al., 2021). Recent studies have shown that some NPCs, such as ellagic acid, gallic acid, methyl gallate, quercetin, chlorogenic acid, protocatechuic acid, flavonoids, tannins, and cyanidin, regulate intestinal microbiota and thus promote animal health. Supplementation of ellagic acid regulates the disorder of cecal microflora caused by paraquat (PQ) stimulation (Qi et al., 2022). Red-osier dogwood polyphenols, rich in gallic acid, methyl gallate, ellagic acid, and quercetin, increase *Bacillus* and *Lactobacillales* in the ileum (Zheng et al., 2021). Likewise, chlorogenic acid increases cecum *Bacillus* and *Bifidobacterium*, increases colon *Lactobacillus*, and reduces colon *Escherichia coli* (*E. coli*) in piglets (Zhang et al., 2018b). The extracts from *Broussonetia papyrifera* leaf and rosemary were reported to alleviate the diarrhea of piglets by regulating the intestinal microbiota (Chen et al., 2020; Yang et al., 2021b). A study has reported that protocatechuic acid effectively increases the ratio of *Firmicutes/Bacteroidetes* in the cecum of piglets (Hu et al., 2020), consistent with the results in a poultry study where eucalyptus leaf polyphenols (rich in flavonoids and tannins) are supplemented in feed (Li et al., 2020). Supplementing grape skin extract (rich in cyanidin) has been reported to reduce ileal *E. coli* in chickens' intestines (Nardoia et al., 2020). Blueberry and cranberry extract (rich in cyanidin) in mouse diet increases the abundance of *Firmicutes* but decreases the abundance of *Bacteroidetes* (Liu et al., 2021b). Similarly, pomegranate peel polyphenol (rich in ellagic acid) can reverse the high proportion of *Firmicutes* and *Bacteroidetes* encountered in high-

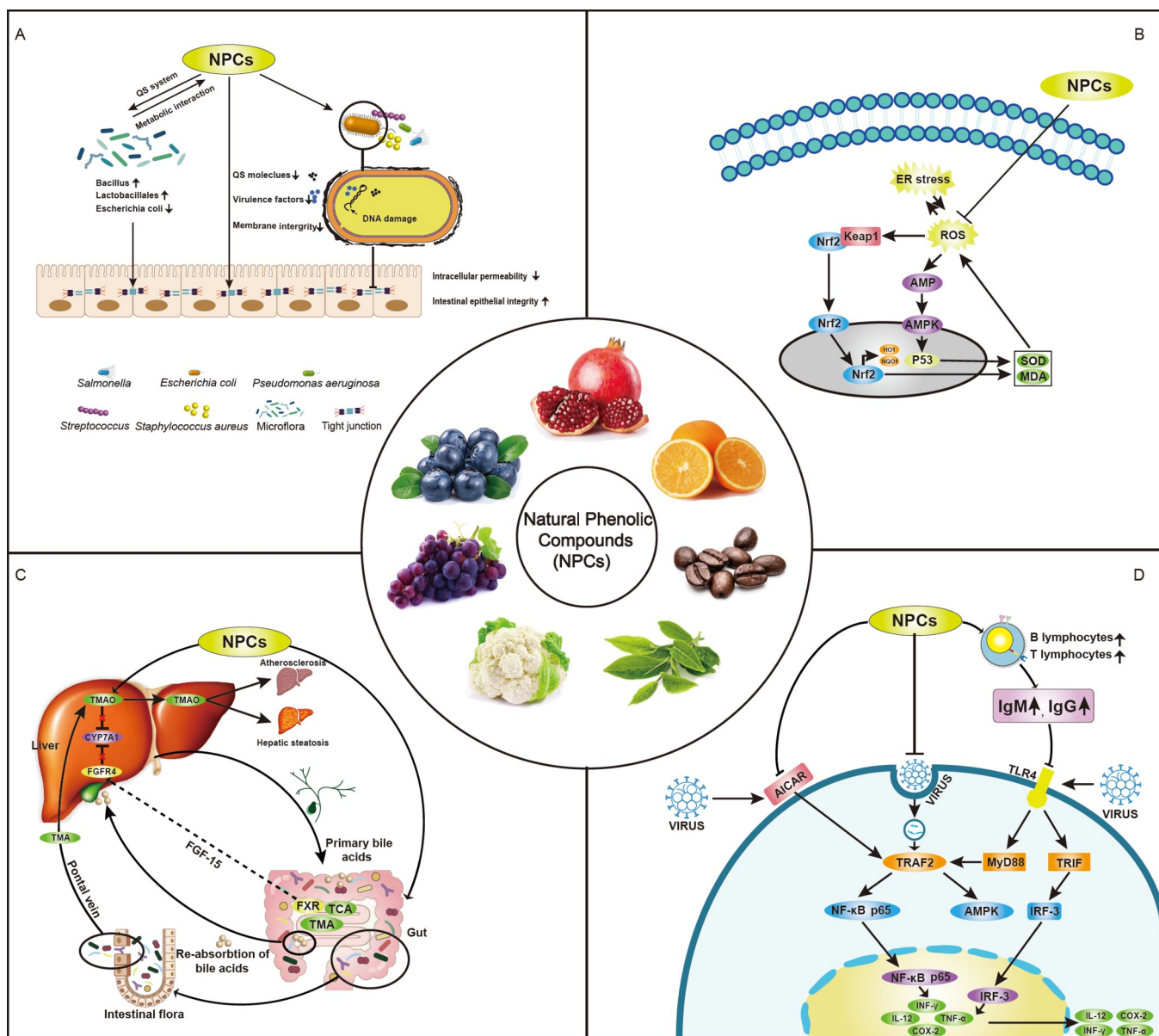


Figure 6 Bioactive properties of NPCs in reducing livestock disease. A, NPCs interfere with pathogenic bacteria, support intestinal microflora, and bolster the intestinal epithelial barrier. B, The mechanisms underlying NPCs' regulation of oxidative stress. C, NPCs regulate intestinal microflora and bile acids through the gut-liver axis. D, NPCs intervene in immunomodulatory and antiviral mechanisms. Abbreviations: AMP, adenosine monophosphate-activated protein; AMPK, adenosine monophosphate-activated protein kinase; HO1, Haem oxygenase-1; IgM, immunoglobulin M; IgG, immunoglobulin G; IFN- γ , interferon-gamma; Keap1, Kelch-like-ECH-associated protein 1; MDA, malondialdehyde; NQO1, NAD(P)H: quinone oxidoreductase 1; SOD, superoxide dismutase; TNF- α , tumor necrosis factor- α . TMA, trimethylamine; TMAO, trimethylamine-N-Oxide; FXR, farnesoid X receptor; TGA, transglutaminase; TGR5, G protein-coupled bile acid receptor-5; TLR4, Toll-like receptor 4; AICAR, 5-aminoimidazole-4-carboxamide-1- β -D-ribofuranoside; TRIF, Toll-interleukin 1 receptor domain-containing adapter inducing interferon- β ; COX-2, cyclooxygenase-2; TRAF-2, TNF receptor-associated factor-2; MyD88, myeloid differentiation primary response 88; FGF-19/15, fibroblast growth factor-19/15.

fat diet-fed rats (Zhao et al., 2019).

The mechanisms of NPCs in regulating intestinal ecology vary widely but generally exert their effects via the following mechanisms: (i) disturbing virulence factors, (ii) bacterial membrane disruption, (iii) inactivating intracellular enzymes, (iv) causing oxidative stress, and (v) inducing DNA damage of specific microbiota (Renzetti et al., 2020). Additionally, NPCs affect the defensive mechanisms and metabolic pathways of the bacteria, such as amino acid

metabolism, protein synthesis, and carbon cycling. Conversely, bacteria can transform NPCs into beneficial substances and modify gut milieu. Such interactions among NPCs, internal milieu, and bacteria also indirectly affect intestinal ecology. NPCs can also affect bacterial quorum sensing (QS) systems via modulation of acylated homoserine lactones in Gram-negative bacteria and oligopeptides in Gram-positive bacteria (Nazareth et al., 2021). For instance, green tea polyphenols were reported to interfere with the

production of QS signaling molecule by pathogens, inhibiting bacterial growth (Zhu et al., 2015). Although studies have reported similar results, particular NPC's specific mechanisms of action on intestinal bacterial functions remain unclear, which need to be furtherly investigated to promote informed and effective use of these agents.

NPCs protect against oxidative stress

Oxidative stress causes biological damage, which in turn affects growth performance, intestinal development, and immune homeostasis in animals (Guo et al., 2020b). Numerous studies suggest that NPCs (e.g., ellagic acid, tea polyphenols, protocatechuic acid, resveratrol) are beneficial in regulating antioxidant-related enzyme activity and preventing over-production of reactive oxygen species (ROS). For example, ellagic acid increases serum superoxide dismutase activity, reduces serum malondialdehyde, promotes nuclear translocation of nuclear factor erythroid 2-Related Factor 2 (Nrf2), and enhances the ratio of heme oxygenase-1 (HO-1) and NAD(P)H to quinone oxidoreductase 1 (NQO1) protein in the small intestinal mucosa, by which protects the PQ-stimulated weaned piglets against oxidative damage (Xiao et al., 2022). Similarly, ellagic acid was reported to improve antioxidant capacity of heat-stressed broilers (Yang et al., 2022). Evidence shows that protocatechuic acid ameliorates oxidative damage in human umbilical vein endothelial cells by decreasing intracellular ROS and malondialdehyde levels and enhancing endogenous antioxidant enzyme activity. Protocatechuic acid also up-regulates the phosphorylation level of AMP-activated protein kinase (AMPK) by activating liver kinase B1 and promoting the expression of p-Nrf2 and HO-1 (Han et al., 2018). Resveratrol in red wine has been reported to remove oxygen radicals and prevent lipid peroxidation (Gorelik et al., 2008) and protect bovine mammary epithelial cells from H₂O₂-induced oxidative stress via the Keap-Nrf2 pathway (Jin et al., 2016). The extracts from red-osier dogwood, which consists of high levels of phenolic compounds, were revealed to protect the intestinal cells against oxidative damage and inflammation (Jiang et al., 2019; Yang et al., 2019). Moreover, resveratrol treatment partially eliminates the adverse effects on coenzyme Q9 and vitamin E levels caused by UV-B radiation in rat hepatocytes (Cavallini et al., 2016).

NPCs reduce pathogenic bacteria

Pathogenic bacteria are the primary cause of diarrhea, gastrointestinal infections, and food poisoning. Recent studies have shown that NPCs (e.g., pyrogallol, resveratrol, rutin, quercetin, catechin, protocatechuic acid, gallic acid, ellagic acid, EGCG, and curcumin) have broad-spectrum anti-

bacterial activity, including QS system inhibition, reducing bacterial growth, and impeding bacterial polysaccharides synthesis. Specifically, pyrogallol inhibits host cell invasion and intracellular survival of *Salmonella enterica* by down-regulating QS, invasion, virulence, and efflux pump genes (Birhanu et al., 2021). Similarly, olive oil polyphenol extract can completely inhibit growth and cell damage by *Salmonella* and *Staphylococcus aureus*, accompanied by reduced intracellular ATP, inhibited cell membrane depolarization, and lower bacterial protein content and cytoplasm leakage (Guo et al., 2020a). Furthermore, resveratrol inhibits biofilm formation by *Pseudomonas aeruginosa* PAO1 and avian pathogenic *E. coli* by inhibiting QS (Vasavi et al., 2017) and affecting proteins abundance, especially for chemotaxis proteins in two-component systems (Ruan et al., 2021). Evidence suggests that rutin can inhibit the growth and biofilm formation of *Pseudomonas aeruginosa* by reducing bacterial cell viability, exopolysaccharide, and extracellular DNA levels (Ivanov et al., 2022). Quercetin extracted from sugarcane bagasse has been reported to inhibit the growth of *Staphylococcus aureus*, *Listeria Monocytogenes*, *E. coli*, and *Salmonella typhimurium* by destroying their cellular proteins (Zhao et al., 2015). *In vivo* and *in vitro* experiments have shown that catechin inhibits the formation of *Stenotrophomonas maltophilia* biofilms as well as the colonization and infection of *Stenotrophomonas maltophilia* in cystic fibrosis mice (Vidigal et al., 2014). The antibacterial effect of tea polyphenols was demonstrated in a study that used the broth microdilution method to show the number of *ETEC* K88 adhering to IPEC-J2 cells in the polyphenol-treated group was significantly lower than that in the control group (Ma et al., 2021). Protocatechuic acid, gallic acid, ellagic acid, and resveratrol have all been reported to inhibit the growth of *Pseudomonas aeruginosa* (Jayaraman et al., 2010; Zhou et al., 2018). Green tea-derived EGCG attenuates *Staphylococcal* alpha-hemolysin (Hla)-induced NLRP3 inflammasome activation via ROS-mediated mitogen-activated protein kinase (MAPK) pathways (Liu et al., 2021a). Furthermore, EGCG inhibits the type III secretion system of enteropathogenic and enterohemorrhagic *E. coli*, *Salmonella enterica* Serovar Typhimurium, and *Yersinia pseudotuberculosis* via inhibition of cell invasion and adhesion (Nakasone et al., 2017). Similarly, EGCG can inhibit *Streptococcus* biofilm formation by suppressing *Gtf* gene expression (Xu et al., 2012). Tea catechins reduce *Streptococci*-induced inflammation by modulating MAPK pathways in Toll-like receptor 2 ligand-stimulated cells (Hirao et al., 2010). Pretreatment with resveratrol reduces pneumoniae infection in rats by inhibiting IκB phosphorylation and subsequent NF-κB activation (Lu et al., 2008). Moreover, curcumin decreases *V. vulnificus*-induced NF-κB translocation in cells, thus protecting mice from septicemia (Na et al., 2011).

Antiviral NPCs

African swine fever virus (ASFV), porcine reproductive and respiratory syndrome virus (PRRSV), and avian influenza virus (AIV, H9N2, H3N2) cause death, hemorrhagic fever, and reproductive disorders in livestock and poultry (Zhang et al., 2021). Studies have demonstrated that the NPCs resveratrol, procyanidins, wogonin, cirsimaritin, and flavonoids inhibit cellular virus adhesion, interfere with virus proliferation, and block viral nucleic acid replication. In Madin-Darby canine kidney and human lung epithelial cells, wogonin inhibits influenza virus replication in a time-dependent manner by regulating AMPK activation (Seong et al., 2018). Cirsimaritin reduces viral titers and influenza virus RNA and protein synthesis in a dose-dependent manner by inactivating the NF- κ B/p65 signal pathway, down-regulating classic proinflammatory cytokines (TNF- α , IL-1 β , IL-8, and IL-10) and the inflammation-related protein COX-2 (Yan et al., 2018). Tea polyphenol inhibits PRRSV infection in Marc-145 cells by preventing p65 entry into the nucleus to suppress the activation of the NF- κ B signaling pathway (Wang et al., 2021b). Resveratrol and oxidized resveratrol inhibit ASFV DNA replication, protein synthesis, and viral replication site synthesis in monkey kidney cells in a dose-dependent manner (Galindo et al., 2011). Furthermore, by upregulating sterile alpha and armadillo motif protein, resveratrol inhibits the Toll-interleukin 1 receptor domain-containing adapter inducing interferon- β (TRIF)-dependent pathway contributing to anti-inflammatory pathways after viral infection (Liu et al., 2014). Procyanidins exhibit potent antiviral activity against PRRSV infection in Marc-145 cells by inhibiting viral RNA synthesis, protein expression, and viral production (Zhang et al., 2018a). In fertilized eggs infected with AIV H9N2, acacia leaf extract (flavonoids) blocks viral replication (Ghoke et al., 2018).

NPCs regulate immunity

Many studies support immune-regulatory capabilities of NPCs (e.g., resveratrol, quercetin, and anthocyanins) in animals. A study showed that resveratrol in feed increases T lymphocytes, enhances the efficacy of vaccination to produce antibodies against swine fever virus and foot-and-mouth virus in piglets, and increases IgG and IgM levels in the serum. The humoral immune response, reflected by IFN- γ and TFN- α expression levels, could be regulated by resveratrol supplementation, supporting its use as an adjuvant to enhance vaccine efficacy in piglets (Fu et al., 2018). Inhibition of NF- κ B and the NF- κ B-driven gene IL-2 by epicatechin and catechin reduces the production of phorbol 12-myristate 13-acetate (PMA)-triggered oxidants in T cells (Mackenzie et al., 2004). Furthermore, resveratrol can enhance the cell-mediated immune response and block the Toll-

like receptor 4 (TLR4) signal pathway (Yusuf et al., 2009). Quercetin supplementation rescues splenocyte loss in irradiated mice (Jung et al., 2012). Cassia anthocyanidins increase T and B cell counts and reduce ROS production in response to *E. coli* infection, thus enhancing immune regulation in aged rats (John et al., 2011).

NPCs improve intestinal epithelial barrier

The intestinal epithelial barrier separates animal internal circulation from the intestinal environment and prevents pathogenic bacterial invasion. A variety of NPCs, including puerarin, ellagic acid, protocatechuic acid, resveratrol, anthocyanins, procyanidins, magnolol, and hesperidin, have been reported to improve intestinal epithelial barrier function and reduce and/or repair intestinal damage by regulating intestinal tight junction protein expression and paracellular permeability. For instance, puerarin helps rebuild the mucus layer and regulate mucin-utilizing bacteria in rats (Wu et al., 2020). Our previous study in PQ-challenged piglets revealed that ellagic acid increases the expression of intestinal tight junction proteins claudin-3 and occludin, suggesting that ellagic acid could alleviate intestinal barrier damage caused by PQ stimulation (Xiao et al., 2022). Similarly, ellagic acid enhances the expression of ZO-1, claudin-1, and claudin-4 by interfering with the Nrf-2 pathway in mice (Singh et al., 2019). Protocatechuic acid has been reported to enhance the ZO-1 and claudin 1 expression in the intestinal mucosa of piglets (Hu et al., 2020). Resveratrol treatment improves intestinal barrier function and increases ZO-1, claudin 1, and occluding levels in H₂O₂-stressed IPEC-J2 cells. Furthermore, resveratrol enhances intestinal barrier through the regulation of the P13K/Akt-mediated Nrf2 signaling pathway (Zhuang et al., 2019). Likewise, purple sweet potato extract (rich in anthocyanin) improves intestinal barrier function in Caco-2 cells, by increasing the trans-epithelial electrical resistance value, reducing paracellular permeability, and increasing the expression levels of ZO-1, occludin, claudin-1, and claudin-2 (Sun et al., 2018). Anthocyanidins and procyanidin in blueberries reverse the pathogenic reduction of trans-epithelial electrical resistance value caused by *E. coli* infection in Caco-2 cells in a dose-dependent manner (Polewski et al., 2020). Resveratrol supplementation in chicken and duck diets increases the expression levels of ZO-1, claudin-1, and occludin, thereby improving intestinal barrier function (Yang et al., 2021a; Zhang et al., 2017). Magnolol also increases the protein level of occludin in the ileum of chickens (Chen et al., 2021) and hesperidin improves the protein levels of ZO-1 and occludin in Caco-2 cells. In Caco-2 and RAW264.7 cell co-culture, hesperidin treatment is capable of preventing DSS- and LPS-induced barrier damage (Zhang et al., 2020).

NPCs regulate the gut-liver axis

Gut microbiota plays a key role in the bidirectional interaction between the gut and liver, called the gut-liver axis. Plenty of evidence shows that NPCs including resveratrol, flavonoids, and epigallocatechin-3-gallate aid in regulating intestinal and liver lipid metabolism, gut barrier function, and gut microbiota. For example, resveratrol attenuates trimethylamine-N-oxide (TMAO)-induced atherosclerosis (AS) by regulating TMAO and bile acids (BAs) biosynthesis via remodeling gut microbiota, which in turn regulates the gut-liver axis (Chen et al., 2016). BA neosynthesis is partially mediated through the FXR-FGF15 axis, in which the resveratrol might be involved in cholesterol 7 α -hydroxylase (CYP7A1) expression and hepatic BA neosynthesis (Chen et al., 2016). Bisphenol A ingestion by maternal rats results in oxidative damage to maternal rat liver and abnormal gut microbiota distribution in their offspring, but was inhibited by resveratrol's ability to modulate antioxidant capacity and rescue gut microbiota (Liao et al., 2021). Coffee is rich in flavonoids and helps prevent high-fat diet-induced non-alcoholic fatty liver diseases in mice by reducing hepatic fat deposition and metabolic derangement through improving pathways underlying liver fat oxidation, intestinal cholesterol efflux, energy metabolism, gut microbiota, and gut permeability (Vitaglione et al., 2019). Evidence shows that the gut-liver axis is also the main target of cranberry polyphenols to reverse the high fat-high sucrose diet-induced hepatic steatosis (Anhê et al., 2017). In some situations, excessive accumulation of visceral and hepatic fat, abnormal liver function, and elevated concentrations of serum and liver inflammatory cytokines can be reversed by epigallocatechin-3-gallate (Huang et al., 2020). In a mouse model of high-fat, high-sucrose diet-induced obesity, polyphenol-rich foods such as alpine bearberry, lingonberry, and cloudberry extracts are able to alleviate intestinal inflammation, insulin resistance, and hepatic steatosis via gut-liver axis regulation (Anhê et al., 2018).

There is still much to learn about the bioactive properties of NPCs and their potential for in-feed antibiotic replacement. For example, most studies only utilize crude NPCs extracts to evaluate their properties *in vitro/in vivo*, but the underlying mechanisms driving NPC-induced improvements in animal health have not been fully clarified. In addition, the biological properties of a vast array of NPCs have not been explored. Therefore, it is essential to pursue high-throughput bioactivity screening for the NPCs and further explore their potential targets and mechanisms to aid in informed and in-depth applications.

Implementations of NPCs as alternatives to in-feed antibiotics

Given the various bioactivities mentioned above, NPCs re-

present promising alternatives to in-feed antibiotics for improving intestinal function, immunity, and overall animal health. In recent years, a considerable number of studies have validated NPCs as alternatives to in-feed antibiotics. In most studies, NPCs, in the form of whole extracts or extracted compounds, have been reported to effectively promote animal health and growth. In this section, we summarize practical instances of NPCs being used as alternatives to in-feed antibiotics.

Studies comparing in-feed NPCs to antibiotics for piglets

Eucommia ulmoides (EU), as a functional plant resource, is widely used in piglet feed due to its high concentrations of chlorogenic acid and flavonoids. For example, research by Yuan et al. demonstrated that the growth performance and diarrhea index of 0.01% EU-supplemented weaned piglets were comparable to the in-feed antibiotic (quinocetone, virginiamycin, and aureomycin) group. Additionally, EU improved intestinal morphology and reduced *E. coli* colonization of weaned piglets (Yuan et al., 2020). The catechins and protocatechuic acid richen in green tea have also been demonstrated as potential alternatives to in-feed antibiotics. Ohno et al. (2013) utilized catechins in dry green tea leaves as an alternative to in-feed antibiotics for piglets and found that 1% dry leaf powder exerts similar growth-promoting effects as in-feed antibiotic treatment (avilamycin and colistin). Recently, Hu et al. (2020) reported that 4,000 mg kg⁻¹ protocatechuic acid supplementation alleviated oxidative stress response induced by LPS, decreased IL-2, IL-6, and TNF- α expression levels, and increased the expression of tight junction proteins ZO-1 and claudin-1. A Chinese herbal medicine mixture composed of *Lonicera maackii*, *As-tragalus membranaceus*, *Folium cortex eucommiae*, and *Codonopsis pilosula* at 1,000 mg kg⁻¹ was observed to improve the intestinal morphology and expression of ileum nutrient transporters SLC6A9, SLC15A1, and SLC5A1 (Wang et al., 2020b).

NPC-rich plant extracts have been widely investigated for in-feed antibiotic replacement. For example, 500 mg kg⁻¹ *Illicium verum* extract or 250 mg kg⁻¹ EU leaf extract administered to nursery piglets enhances growth performance and improves antioxidative and anti-inflammatory parameters in piglets (Jiang et al., 2020). Supplementation of 40 mg kg⁻¹ *Tartary buckwheat* flavonoids in pig diet improved growth, nutrient digestibility, and increases serum IgA and IgM levels (Cui et al., 2019). Long et al. (2019) showed that 200 mg kg⁻¹ *Forsythia suspense* extract in feed reduced diarrhea rate and fecal *E. coli* content, improved average daily weight gain, nutrient digestibility, and the ratio of villus height/crypt depth for nutrient adsorption, thereby improving feeding efficacy of piglets. Feeding with 300 mL t⁻¹ citrus extract improved the serum antioxidant capacity and in-

testinal morphology of the piglets (Cui et al., 2020b). Wang et al. (2021a) showed that 50–200 mg kg⁻¹ of guava leaf extract reduced diarrhea rate, improved antioxidant capacity, enhanced intestinal tight junction protein expression, and benefitted the intestinal morphology of piglets. Grape seed procyanidin trimers (250 mg kg⁻¹) improved growth, mucosal morphology, increased intestinal tight junction protein expression, reduced diarrhea rate, and regulated antioxidant-related enzyme content (Han et al., 2016).

Monomeric NPC compounds, such as resveratrol, curcumin, tannins, and microencapsulated tannic acid, were found to promote the health of piglets. For example, 0.2% resveratrol improved piglet growth, increased nutrient digestibility, enhanced serum IgG level, and reduced the shedding of enterocytes comparably to in-feed antibiotics (apramycin) (Ahmed et al., 2013). Gan et al. (2019) revealed that 300 mg kg⁻¹ curcumin or resveratrol improved intestinal microflora and ameliorated intestinal inflammation in weaned piglets, possibly via down-regulation of TLR4 expression and key inflammatory factors such as IL-1 β and TNF- α . Wang et al. (2020a) reported that microencapsulated tannic acid (1,000 mg kg⁻¹) improved intestinal morphology, enhanced the expression of intestinal amino acid transporters, and positively regulated intestinal microbiota. Administration of 0.3% feed additive (89.3% benzoic acid, 1.8% thymol, 3.2% eugenol, and 5.7% piperine) increased average daily weight gain, improved the digestibility, and reduced the pathogenic bacteria *E. coli* and *Campylobacter* in cecal contents (Silva Júnior et al., 2020).

Essential details and outcomes of these studies including experimental animals, experiment period/ages, experimental groups, dosages, and effects are shown in Table 1.

Studies comparing in-feed NPCs to antibiotics for chickens and ducks

Similar to the findings in piglets, NPC-rich plant extracts promote poultry growth and health. For example, citrus extracts improved growth performance and intestinal morphology, increased most plasma essential amino acids (EAA), and boosted mRNA expression levels of polypeptide and sugar transporters in broilers (Yu et al., 2020). Herawati et al. (2019) reported that 2% mangosteen peel extract increased the growth of broilers absent antibiotic resistance. Chestnut extract (1,000 mg kg⁻¹) increased average daily gain, decreased feed: egg ratio, increased crude protein digestibility, increased serum and tissue total antioxidant capacity, and increased beneficial molecular markers such as glutathione peroxidase, superoxide dismutase, and serum IgG level in the broiler chicken (Liu et al., 2020). *Forsythia suspensa* extract (100 mg kg⁻¹) improved growth, intestinal morphology, calcium digestibility on day 21, energy digestibility on day 42, and decreased the number of *E. coli* in

the cecum of broilers (Han et al., 2012). A *Medicago sativa* L. (Alfalfa) extract (1,000 mg kg⁻¹) was reported to improve antioxidant parameters in broilers (Dong et al., 2011). *Artemisia annua* L. aqueous extract (500–2,000 mg kg⁻¹) promoted growth efficiency similar to 50 mg kg⁻¹ chlortetracycline in broilers (Guo et al., 2020c).

Monomeric NPCs broadly illustrate potential as in-feed antibiotic replacements. Lin et al. (2020) found 100–300 mg kg⁻¹ magnolol or 30 mg kg⁻¹ colistin sulfate (in-feed antibiotics) improved chicken growth performance. Furthermore, magnolol supplementation in feed significantly increased the antioxidant capacity of broiler chickens compared to colistin sulfate. In addition, 300–600 mg kg⁻¹ protocatechuic acid supplementation can improve the growth, antioxidant capacity, intestinal immune function, and intestinal microflora profiles of yellow-feathered broilers (Wang et al., 2019). A study by Reis et al. (2018) revealed that thymol, carvacrol, and cinnamic aldehyde supplementation increased the body weight and meat quality of broilers. Zhu et al. (2019) found that carvacrol and thymol (200–400 mg kg⁻¹) improved the feed conversion ratio. Similarly, Bosetti et al. (2020) reported beneficial effects of 100–400 mg kg⁻¹ microencapsulated carvacrol (30%) and cinnamaldehyde (60%) on the production and meat quality of broilers. Genistein (10–320 mg kg⁻¹) enhanced the growth and immune fortitude of broilers (Rasouli and Jahanian, 2019) and increased the average daily feed intake, feed conversion rate, and lymphocytes count, and decreased the heterophil to lymphocyte ratio. Kusmayadi et al. (2019) reported that raw 1% mangosteen peel and 1% turmeric flour supplementation increased ducks' body weight, reduced mortality, and reduced jejunum *E. coli*.

Essential parameters of these studies in terms of experimental animals, experiment period/ages, experimental groups, dosages, and effects are shown in Table 2.

Studies comparing in-feed NPCs to antibiotics for ruminants

Cinnamaldehyde, biochanin A, and tannin promote the health of ruminants and show other improvements comparable to in-feed antibiotics. Lourenco et al. (2008) reported that 500 mg L⁻¹ cinnamaldehyde increased polyunsaturated fatty acids and butyrate in the rumen with similar efficacy as 12 mg L⁻¹ monensin (in-feed antibiotic). Biochanin A (30%) in a high-fiber diet prevented subacute ruminal acidosis (SARA), decreased propionate, increased total short-chain fatty acid yield, and inhibited the growth of Gram-positive coccus. Additionally, biochanin A ameliorated the reduced cellulolytic microbes caused by SARA and increased average daily weight gain and acetate content in bulls (Harlow et al., 2021). Hassanat et al. (2013) found that 20–200 mg kg⁻¹ tannin increased total butyric acid, valerate acid contents, gas

Table 1 Trails of NPCs as alternatives to in-feed antibiotics for piglets^{a)}

NPCs	In-feed antibiotics	Animals (breed)	Experiment period/ages	Experimental groups and dosages	Effects	Ref.
Flavonoids (in <i>Eucommia ulmoides</i>)	Quinocetone, virginiamycin, and aureomycin	480 piglets (Duroc×Landrace×Large Yorkshire)	25-d-old to 60-d-old	Basal diet, basal diet+antibiotics (75 mg kg ⁻¹ quinocetone, 20 mg kg ⁻¹ virginomycin, 50 mg kg ⁻¹ aureomycin), basal diet+0.01% <i>Eucommia ulmoides</i> flavonoids	NPC group and antibiotic group: ADG↑, diarrhea index↓; NPC group: villus height/crypt depth↑, villus height in jejunum and ileum↑, ileum <i>Lactobacillus</i> ↑, number of <i>E. coli</i> in the ileum↓	Yuan et al., 2020
Catechin (in dry green tea leaves)	Avilamycin and colistin	26 Tokyo-X pigs	28-d-old to 56-d-old	Basal diet, basal diet+100 mg kg ⁻¹ in-feed antibiotics, basal diet+1% dry green tea leaves	NPC group and antibiotic group: mortality↓, body weight↑	Ohno et al., 2013
Resveratrol	Apramycin	48 piglets (Duroc×Landrace×Large Yorkshire)	28-d-old to 56-d-old	Basal diet, basal diet+0.002% apramycin, basal diet+0.2% resveratrol	NPC group and antibiotic group: ADG↑, ADFI↑, digestibility of nutrients (DM, N) ↑, serum IgG↑, shedding amount of <i>E. coli</i> ↓	Ahmed et al., 2013
Curcumin and resveratrol	Olaquinox, kitasamycin, and chlortetracycline	180 piglets (Duroc×Large White×Landrace)	21-d-old to 49-d-old	Basal diet, basal diet+300 mg kg ⁻¹ antibiotics (159 mg kg ⁻¹ olaquinox+81 mg kg ⁻¹ kitasamycin+60 mg kg ⁻¹ chlortetracycline), basal diet+curcumin and resveratrol (300 mg kg ⁻¹ curcumin+resveratrol, 100 mg kg ⁻¹ curcumin+resveratrol, 300 mg kg ⁻¹ curcumin, 300 mg kg ⁻¹ resveratrol)	Curcumin group and antibiotic group: <i>E. coli</i> ↓; curcumin-resveratrol group and antibiotic group: serum IL-1β and TNF-α↓, serum IL-10↑, serum IgG↑, total bacterial copy in jejunum and ileum↓, <i>E. coli</i> in jejunum and ileum↓; NPC group: serum IL-1β and TNF-α↓, serum IL-10↑, ileum and serum IgA↑	Gan et al., 2019
Flavonoids (in <i>Tartary buckwheat</i>)	Colistin sulfate	50 piglets (Large White×Landrace)	35-d-old to 63-d-old	Basal diet, basal diet+40 mg kg ⁻¹ colistin sulfate, basal diet+40 mg kg ⁻¹ <i>Tartary buckwheat</i> flavonoids	NPC group and antibiotic group: ADG↑, nutrient digestibility↑; NPC group: serum IgA and IgM↑	Cui et al., 2019
Protocatechuic acid	Aureomycin	36 piglets (Pig Improvement Company line 337×C48)	28-d-old to 49-d-old	Basal diet, basal diet+50 mg kg ⁻¹ aureomycin, basal diet+4,000 mg kg ⁻¹ protocatechuic acid	NPC group and antibiotic group: tight junction proteins (ZO-1 and claudin-1)↑, TBARS↓, IL-6↓; NPC group: GSH-px↑, serum IL-2 and TNF-α↓	Hu et al., 2020
<i>Illicium verum</i> extract, <i>Eucommia ulmoides</i> leaf extract	Chlortetracycline	48 DLY piglets (Duroc×Landrace×Yorkshire) and 48 LCB piglets (Chinese native Licha-black)	35-d-old to 84-d-old and 42-d-old to 91-d-old	Basal diet, basal diet+50 mg kg ⁻¹ chlortetracycline, basal diet+500 mg kg ⁻¹ <i>Illicium verum</i> extract, basal diet+250 mg kg ⁻¹ <i>Eucommia ulmoides</i> leaf extract	NPC group and antibiotic group: ADG↑; NPC group: serum and liver T-SOD↑, GSH-Px↑, TNF-α↓, Nrf2↑	Jiang et al., 2020
<i>Forsythia suspense</i> extract	Chlortetracycline	108 Duroc×(Landrace×Yorkshire) weaned piglets	28-d-old to 63-d-old	Basal diet, basal diet+75 mg kg ⁻¹ chlortetracycline, basal diet+200 mg kg ⁻¹ <i>Forsythia suspense</i> extract	NPC group and antibiotic group: diarrhea rate↓; NPC group: ADG↑, nutrient digestibility↑, fecal nitrogen production↓, number of <i>E. coli</i> in fecal↓, villus height/crypt depth↑	Long et al., 2019
Citrus extract	Chlortetracycline	144 weaned piglets (Duroc×Landrace×Large White)	28-d-old to 56-d-old	Basal diet, basal diet+75 g t ⁻¹ chlortetracycline, basal diet+300 mL t ⁻¹ citrus extract	NPC group and antibiotic group: albumin↑, crypt depth (duodenum, jejunum, and ileum)↓, villus height/crypt depth (ileum)↑; NPC group: plasma albumin↑, duodenal alkaline phosphatase and lipase activity↑, jejunum alkaline phosphatase and trypsin activity↑	Cui et al., 2020b
Tannic acid	Chlortetracycline, enramycin, and oxytetracycline calcium	45 piglets (Duroc×Landrace×Yorkshire)	21-d-old to 35-d-old	Basal diet, basal diet+antibiotics (375 mg kg ⁻¹ chlortetracycline, 20%, 500 mg kg ⁻¹ enramycin, 4%, 1,500 mg kg ⁻¹ oxytetracycline calcium, 20%), basal diet+tannic acid (500, 1,000, and 1,500 mg kg ⁻¹)	NPC group and antibiotic group: duodenal crypt depth↓, villus height in duodena↑, intestinal amino acid transporters expression (threonine)↑	Wang et al., 2020a

(To be continued on the next page)

(Continued)

NPCs	In-feed antibiotics	Animals (breed)	Experiment period/ages	Experimental groups and dosages	Effects	Ref.
Herbal extract mixture (<i>Lonicera maackii</i> , <i>Astragalus membranaceus</i> , <i>Folium cortex eucommiae</i> , and <i>Codonopsis pilosula</i>)	Chlortetracycline, enramycin, oxytetracycline calcium	27 piglets (Duroc×Landrace×Yorkshire)	21-d-old to 35-d-old	Basal diet, basal diet+antibiotics (375 mg kg ⁻¹ chlortetracycline, 20%, 500 mg kg ⁻¹ enramycin, 4%, 1,500 mg kg ⁻¹ oxytetracycline calcium, 50%), basal diet+1,000 mg kg ⁻¹ herbal medicine mixture	NPC group and antibiotic group: crypt depth of duodenum↓, villus height/crypt depth of duodenum and jejunum↑; NPC group: ileal maltase activity↓, the expression of ileum nutrient transporters (SLC6A9, SLC15A1, and SLC5A1)↑	Wang et al., 2020b
NPC mixture (benzoic acid, 89.3%, thymol, 1.8%, eugenol, 3.2%, and piperine, 5.7%)	Colistin	108 crossbred barrows	21-d-old to 65-d-old	Basal diet, basal diet+40 mg kg ⁻¹ colistin, basal diet+0.3% feed additive	NPC group and antibiotic group: nutrient digestibility↑; NPC group: ADG↑, ADFI↑, number of <i>E. coli</i> in cecal↓, <i>Campylobacter</i> pathogenic bacteria↓	Silva Júnior et al., 2020
Guava leaf extract	Quinocetone	60 weaned piglets (Duroc ×Yorkshire×Landrace)	21-d-old to 49-d-old	Basal diet, basal diet+50 mg kg ⁻¹ quinocetone, basal diet+guava leaf extract (50, 100, and 200 mg kg ⁻¹)	NPC group and antibiotic group: diarrhea rate↓, serum SOD↑, intestinal tight junction proteins (ZO-1, claudin-1, and occludin) ↑, MDA↓, damage of intestinal mucosa↓; NPC group: villus height↑, villus height/crypt depth↑, jejunum crypt depth↓	Wang et al., 2021a
Proanthocyanidins (in grape seed)	Colistin	120 piglets (Duroc×Landrace×Large White)	28-d-old to 56-d-old	Basal diet, basal diet+250 mg kg ⁻¹ colistin, basal diet+250 mg kg ⁻¹ grape seed, basal diet+250 mg kg ⁻¹ grape seed and half dose antibiotics	NPC group and antibiotic group: ADG↑, ADFI/ADG↑, diarrhea rate↓, intestinal permeability↓, jejunum and ileum villus height↑; NPC group: GPR41 and occludin in the intestinal mucosa↑, SOD↑, GSH-Px↑, GSH↑, MDA↓	Han et al., 2016

a) ADFI, average daily feed intake; ADG, average daily gain; CAT, catalase; DM, dry matter; GSH, glutathione; GSH-Px, glutathione peroxidase; MDA, malondialdehyde; N, nitrogen; SOD, superoxide dismutase; T-AOC, total antioxidant capacity; TBARS, thiobarbituric acid reactive substances; T-SOD, total-superoxide dismutase.

production (CH₄, NH₃), and acetic acid/propionic acid.

NPC-rich plants were also been tested in ruminant feeds and found to be widely beneficial. For example, 20% flax-seed shell supplementation during late lactation reduced the content of β-glucuronidase in rumen fluid and increased the concentration of enterolactone in milk similarly to 16 mg kg⁻¹ monensin (Petit et al., 2009). A mixture of cashew and castor oil (500 mg kg⁻¹) increased propionic acid content in rumen fluid, enhanced milk yield and increased protein, lactose content, and N excretion in milk, but reduced blood urea content in dairy cows (Ferreira de Jesus et al., 2016).

Essential parameters of these studies such as experimental animals, experiment period, experimental groups, dosages, and effects are listed in Table 3.

Conclusion and perspectives

Currently, a variety of NPCs exhibit beneficial effects on animal growth, nutrient digestion, immunity, and intestinal health. NPCs such as chlorogenic acid and curcumin have already been implemented as alternatives to in-feed antibiotics. Compared with other potential alternatives (probiotics, prebiotics, enzyme preparations, acidifiers,

antimicrobial peptides, bacteriophages), NPCs garner many advantages including their natural origin, wide availability, low cost, and multiple mechanistic targets. We propose the following items for consideration in promoting NPC application in livestock farming.

Different forms of NPCs for animal feed

Studies comparing NPCs to in-feed antibiotics, utilized various preparations (air-dried plants, raw extracts, or purified chemical compounds) of NPCs. Generally, the cost of using air-dried plants is lower than both raw extracts and purified chemical compounds. Currently, 29 NPC-rich plants, such as mulberry leaf, magnolia officinalis, and sunflower seed hull, were authorized for feed supplementation in China. Yet uncharacterized compounds in plants may affect the efficacy of the NPCs as alternatives to in-feed antibiotics in unpredictable ways. Therefore, during their implementation in animal feed, the following should be considered in sequence: (i) whether the NPC-rich plants contain sufficient protein and carbohydrate for the replacement of traditional feed resources, (ii) whether antinutritional compounds (such as mycotoxins, tannin, phytic acid, thioglycoside) exist in the NPC-rich plants, (iii) whether such compounds impede microbial or enzyme fermentation for

Table 2 Trails of NPCs as alternatives to in-feed antibiotics for poultry^{a)}

NPCs	In-feed antibiotics	Animals (breed)	Experiment period/ages	Experimental groups and dosages	Effects	Ref.
<i>Artemisia annua</i> L. aqueous extract	Chlortetracycline	240 mixed-sex Arbor Acres broil	1-d-old to 42-d-old	Basal diet, basal diet+50 mg kg ⁻¹ chlortetracycline, basal diet+ <i>Artemisia annua</i> L. aqueous extract (500, 1,000, 1,500, and 2,000 mg kg ⁻¹)	NPC group and antibiotic group: serum, liver, and spleen SOD↑, GSH-Px↑, T-AOC↑, CAT↑, MDA↓; NPC group: ADG↑, FCR↑, ADFI↑	Guo et al., 2020c
Magnolol	Colistin sulphate	275 female Linwu ducks	49-d-old to 70-d-old	Basal diet, basal diet+30 mg kg ⁻¹ colistin sulphate, basal diet+ magnolol (100, 200, and 300 mg kg ⁻¹)	NPC group and antibiotic group: ADG↑, drip loss↓; NPC group: serum SOD↑, T-AOC↑, CAT↑, GSH↑	Lin et al., 2020
Protocatechuic acid	Enramycin	720 male Chinese yellow-feathered broilers.	1-d-old to 52-d-old	Basal diet, basal diet+200 mg kg ⁻¹ enramycin, basal diet+ protocatechuic acid (300 and 600 mg kg ⁻¹)	NPC group and antibiotic group: LW↑, carcass weight↑, spleen index↑, liver MDA↓, jejunum mucosa T-AOC↑, GSH-Px↑; NPC group: ADG↑, F:G↓, ALP↓, DAO↓, liver T-AOC↑, xanthine oxidase activity↓, jejunum mucosa, and plasma IL-10 and TGF-β↑, IFN-γ↓, intestinal barrier tight junction proteins (ZO-1, claudin, and occludin)↑	Wang et al., 2019
Mangosteen peel, turmeric flour	Bacitracin	84 male Cihateup ducks	84-d-old to 140-d-old	Basal diet, basal diet+50 mg kg ⁻¹ bacitracin, basal diet+2% mangosteen peel, basal diet+2% turmeric flour, basal diet+ mangosteen peel (0.5%, 1%, and 1.5%)+turmeric flour (0.5%, 1%, and 1.5%)	NPC group: weight gain↑, mortality↓, number of <i>E. coli</i> in jejunum↓	Kusmayadi et al., 2019
Phytogenic feed additive (thymol, carvacrol, and cinnamic aldehyde)	Zinc bacitracin	240 male broiler chicks	1-d-old to 42-d-old	Basal diet, basal diet+zinc bacitracin, basal diet+phytogenic feed additive (0.5% and 1%)	NPC group: LW↑, chicken quality↑	Reis et al., 2018
Citrus extract	Zinc bacitracin	540 male yellow-feathered broilers	1-d-old to 63-d-old	Basal diet, basal diet+10 mg kg ⁻¹ zinc bacitracin, basal diet+ 10 mg kg ⁻¹ citrus extract	NPC group: ADG↑, F:G↓, jejunum and ileum crypt depth↓, villus height/crypt depth↑, most plasma EAA↑, mRNA expression levels of polypeptide and sugar transporters↑	Yu et al., 2020
Plant extracts (carvacrol, 20%, thymol, 25%)	Virginiamycin	288 male Cobb broiler chickens	1-d-old to 42-d-old	Basal diet, basal diet+30 mg kg ⁻¹ virginiamycin, basal diet+plant extracts (200 and 400 mg kg ⁻¹)	NPC group and antibiotic group: F:G↓	Zhu et al., 2019
Feed additive (carvacrol, 30%, cinnamaldehyde, 60%)	Virginiamycin	600 male chicks	1-d-old to 41-d-old	Basal diet, basal diet+30 mg kg ⁻¹ virginiamycin, basal diet+feed additive (100, 200, and 400 mg kg ⁻¹)	NPC group and antibiotic group: carcass weight↑; feed additive group: chicken quality↑	Bosetti et al., 2020
Mangosteen peel extract	Colistin	30 broiler chicks (Cobb500)	1-d-old to 30-d-old	Basal diet, basal diet+300 mg kg ⁻¹ colistin, basal diet+2% mangosteen peel extract	NPC group and antibiotic group: the third-week LWG↑, FCR↑	Herawati et al., 2019
Genistein	Virginiamycin, zinc-bacitracin	240 male broiler chicks (Cobb 500)	1-d-old to 42-d-old	Basal diet, basal diet+antibiotics (20 mg kg ⁻¹ virginiamycin and zinc-bacitracin), basal diet+genistein (10, 20, 40, 80, 160, and 320 mg kg ⁻¹)	NPC group and antibiotic group: ADFI↑, number of <i>Salmonella</i> and <i>E. coli</i> ↓, villus height↑, crypt depth↑, digestibility of nutrients (crude protein, crude fat, and energy)↑	Rasouli and Jahanian, 2019
Chestnut wood extract	Chlortetracycline	168 male Arbor Acres broilers	1-d-old to 42-d-old	Basal diet, basal diet+75 mg kg ⁻¹ chlortetracycline, basal diet+ 1,000 mg kg ⁻¹ chestnut wood extract	NPC group: final body weight↑, ADG↑, F:G↓, CP digestibility↑, meat quality↑, chicken tissue and serum T-AOC↑, GSH-Px↑, SOD↑, serum IgG↑	Liu et al., 2020
<i>Forsythia suspensa</i> extract	Chlortetracycline	252 male Arbor Acres broilers	1-d-old to 42-d-old	Basal diet, basal diet+chlortetracycline (1–21 days, 80 mg kg ⁻¹ , 22–42 days, 50 mg kg ⁻¹), basal diet+100 mg kg ⁻¹ <i>Forsythia suspensa</i> extract	NPC group and antibiotic group: ADG↑, ADFI↑, apparent digestibility of calcium↑, feed energy↑, number of <i>E. coli</i> in cecum↓; NPC group: villus height in duodenum and ileum↑, villus height/crypt depth↑	Han et al., 2012
Alfalfa extract	Chlortetracycline	360 male Arbor Acres broilers	1-d-old to 42-d-old	Basal diet, basal diet+0.15 g kg ⁻¹ chlortetracycline, basal diet+ 1 g kg ⁻¹ alfalfa extract	NPC group and antibiotic group: ADG↑; NPC group: liver and serum T-SOD↑, GSH-Px↑, SOD↑, serum MDA↓	Dong et al., 2011

a) ADFI, average daily feed intake; ADG, average daily gain; ALP, alkaline phosphatase; CAT, catalase; CP, crude protein; DAO, diamine oxidase; DM, dry matter; FCR, feed conversion ratio; F:G, feed to gain ratio; GSH, glutathione; GSH-Px, glutathione peroxidase; LW, live weight; LWG, live weight gain; MDA, malondialdehyde; N, nitrogen; SOD, superoxide dismutase; T-AOC, total antioxidant capacity; TBARS, thiobarbituric acid reactive substances; T-SOD, total-superoxide dismutase.

Table 3 Trails of NPCs as alternatives to in-feed antibiotics for ruminants^{a)}

NPCs	In-feed antibiotics	Animals (breed)	Experiment period (d)	Experimental groups and dosages	Effects	Ref.
Cinnamaldehyde	Monensin	Rumen-fistulated Holstein cows	5	Basal diet, basal diet+12 mg L ⁻¹ monensin, basal diet+500 mg L ⁻¹ cinnamaldehyde	NPC group and antibiotic group: total SFA↓, total PUFA↑, total MUFA↓, C18:1 trans-10↑, CLA trans-10↑, C18:1 trans-11↓, C18:1 trans-15↓, C18:2 trans-11↑, C18:2 cis-15↑, C18:2n-6↑, C18:3n-3↑; NPC group: total OBCFA↓, butyrate↑, propionate↓, total VFA↓, C18:1 cis-15↓	Lourenco et al., 2008
Flaxseed hulls	Monensin	Rumen-fistulated multiparous Holstein cows	28	Basal diet, basal diet+16 mg kg ⁻¹ monensin, basal diet+20% flax-seed shell	NPC group and antibiotic group: rumen fluid β-glucuronidase↓; NPC group: enterolactone↑	Petit et al., 2009
Biochanin A	Monensin	Rumen-fistulated Holstein steers	29	High fiber diet, high fiber diet +SARA infection, high fiber diet+200 mg monensin (SARA infection), high fiber diet+30% biochanin A (SARA infection)	NPC group (SARA infection) and antibiotic group: rumen fluid PH↑, propionate↓, total SCFA↓, number of starch-degrading bacteria↓, DMD↑, number of 2-DG resistant cellulolytic bacteria↑, fiber degradation↑, GPC↓, amino acid/peptide-fermenting bacteria↓, digestibility, and utilization of high and low-quality hay↑; NPC group (SARA infection): acetate↑, ADG↑	Harlow et al., 2021
Tannins	Monensin	Rumen-fistulated lactating cows	28	Basal diet, basal diet+10 mg L ⁻¹ monensin, basal diet+tannins (20, 50, 100, 150, and 200 g kg ⁻¹)	NPC group and antibiotic group: gas production rate↓, CH ₄ ↓, NH ₃ ↓, acetate/propionate↓, VFA/CH ₄ ↑, branched VFA↓; NPC group: rumen fluid PH↑, total VFA↓, butyrate↓, valerate↓	Hassanat and Benchaar, 2013
Cashew nut shell liquid and castor oil mixture	Monensin	Rumen-fistulated Holstein cows	21	Basal diet, basal diet+22 mg kg ⁻¹ monensin, basal diet+500 mg kg ⁻¹ cashew nut shell liquid and castor oil mixture	NPC group and antibiotic group: propionate↑, milk production↑, lactoprotein↑, lactose↑, N excretion in milk↑; NPC group: blood urea content↓	Ferreira de Jesus et al., 2016

a) ADG, average daily gain; DMD, dry matter disappearance; GPC, Gram-positive cocci; MUFA, monounsaturated fatty acid; N, nitrogen; OBCFA, odd and branched fatty acid; PUFA, polyunsaturated fatty acid; SCFA, short-chain fatty acid; SFA, saturated fatty acid; VFA, volatile fatty acid.

the raw extracts (iv) whether the technology exists to exclude anti-nutritional factors from raw extracts. Importantly, these considerations will better inform the overall costs and efficacy of these different forms of specific NPC-rich plants for in-feed applications.

Utilization of local NPC-rich plants for livestock and poultry

Presently, hybrids of various livestock and poultry species including Duroc×Large White×Landrace pigs, Arbor Acres Broiler chickens, and multiparous Holstein cows are commonly used in most in-feed replacement tests. Superior local species in China, including Ningxiang pigs and Shaziling pigs, have improved meat quality and are well-adapted to local crude feed, and thus exhibit excellent market potential. This is especially true since local plants include NPC-rich species such as *Plantago maxima* in Xinjiang and *Ampelopsis* in Hunan and therefore could be used as antibiotic replacements with such locally adapted livestock and poultry species to provide unique livestock-NPC plant combinations

and farming opportunities. However, further research on this subject is greatly needed to evaluate its potential.

Technology and application standards in NPC production

Standardizing quality control and application of purified NPCs extracts from plants was relatively easy, providing convenient alternatives to in-feed antibiotics. The main challenges faced by the industry are the availability of raw materials, imperfect separation and purification technology, and variable product properties, which increase the cost of NPC purification and application. To solve these issues, further study will be required to understand the synthesis mechanism of NPCs in plants and to improve the breeding of high-yielding and NPC-rich plants. Additionally, *in vitro* synthesis of NPCs by genetically engineering bacteria and yeast may provide a lower-cost method of generating NPCs for animal feed applications. Currently, 13 extracts from NPC-rich plants including *Eucommia ulmoides*, Alfalfa, and *Camellia sinensis* L. were authorized to be used as feed additives in China. To promote the passing of legislation

authorizing further NPC species as feed additives, it will be vital to uncover the mechanisms that allow NPCs to be so effective as alternatives to in-feed antibiotics, optimize the technologies for their production, establish the NPC product standardization procedures, and promote industrial-scale application of NPCs.

Compliance and ethics The author(s) declare that they have no conflict of interest.

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